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Travelling Salesperson Problem (TSP)

Exact Algorithms for TSP

Approximation Algorithm for TSP

Video: Metric TSP and Shortcutting

25 min

Video: Eulerian Walks for approximating TSPs

11 min

Video: Christofides Algorithm and its Analysis

27 min

Lab: Interactive Notes: Approximations for Metric TSPs

1h

Quiz: Approximations for Metric TSPs

Submitted

Heuristics: A Brief Tour

Approximation Schemes and the Knapsack Problem

Problem Set # 4

Approximations for Metric TSPs

Review Learning Objectives

Submit your assignment

Due Mar 3, 11:59 PM IST

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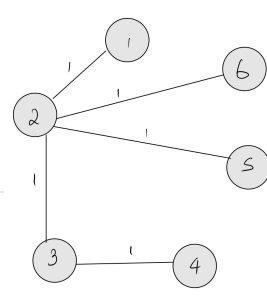
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Suppose we have a TSP instance for which the minimum spanning tree (MST) is shown below.



The edges not in the MST have not been shown. We perform a depth first traversal of the TSP and it yields the following sequence of visits that form a TSP tour:

[1, 2, 5, 6, 3, 4]

Note that we take vertex 1 to be the start/end point.

1. Which of the following are the shortcut edges in the TSP tour constructed? 2 / 2 points

☒ (5,6)

☐ Correct Correct

☐ (1,2)

☒ (6,3)

☐ Correct Correct

☐ (2,5)

☒ (4,1)

☐ Correct Correct

☐ (3,4)

2. What can we say for sure about the cost of the edge $(6, 5)$ denoted $C_{6,5}$ in the original TSP? 2 / 2 points

☒ $C_{6,5} \leq 2$

☐ Correct

Correct: it follows by triangle inequality $C_{6,5} \leq C_{6,2} + C_{2,5}$

☐ $C_{6,5} \geq 1$

☒ $C_{6,5} = C_{5,6}$

☐ Correct

Yes a metric TSP must be symmetric

☐ $C_{6,5} = 2$

3. What can we say for sure about the cost of the TSP tour [1, 2, 5, 6, 3, 4] obtained computed by shortcutting the MST as above? Select all choices that apply 1.5 / 3 points

☐ We have no data since the cost of the edges $C_{5,6}$, $C_{6,3}$ and $C_{4,1}$ are not available. The cost the tour can be arbitrary.

☒ The cost of the tour must be less than or equal to twice the cost of the MST.

☐ Correct

Correct since the tour we construct shortcuts a DFS traversal of the MST, this follows as was explained in the notes.

☒ The cost of the tour must be ≥ 5 , where 5 is the cost of the MST.

☒ This should not be selected

Correct: any TSP tour must have cost greater than the cost of the MST

☐ The cost of the TSP tour must be ≤ 10 .

4. Which of the following statements about the cost of the optimal TSP tour for the instance above can be concluded. 2 / 2 points

☒ The optimal tour cost has to be ≥ 5 .

☐ Correct

True: the optimal TSP tour has to cost as much as the MST.

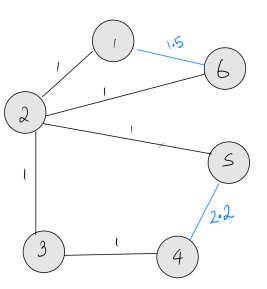
☐ There is insufficient data to place bounds on the optimal tour.

☒ The optimal tour cost has to be ≤ 10 .

☐ Correct

Correct: we already have a tour of cost ≤ 10 . The optimal tour has to do as well or better than that.

Proceeding with the previous example, the odd degree vertices in the MST are $\{1, 4, 5, 6\}$ each of which is a leaf and has degree 1. Suppose we construct a minimum weight matching over these odd degree vertices to obtain the following graph which shows the original MST edges in black and the matching edges and weights in blue.



An Eulerian tour of this graph is obtained as

[1, 2, 5, 4, 3, 2, 6, 1]

You can verify that this tour visits all edges in the graph formed by MST and matching edges.

5. Which of the following is the TSP tour that will be obtained by shortcutting the Eulerian tour? 1 / 1 point

☐ [1, 2, 5, 4, 3, 2]

☐ [1, 2, 3, 4, 5, 6]

☒ [1, 2, 5, 4, 3, 6]

☐ [1, 2, 6, 5, 4, 3]

☐ Correct

Correct

6. Which of the following is a valid observation about the cost of the TSP tour obtained by shortcutting the Eulerian tour? 1.25 / 3 points

☒ The cost of the tour must be less than or equal to the cost of all the edges in the matching + cost of all edges in the MST

☐ Correct

Correct.

☒ Cost of TSP tour is less than or equal to 8.7

☐ Correct

Correct: we can easily show using data given and metric property

$C_{1,2} + C_{2,5} + C_{5,4} + C_{4,3} + C_{3,6} + C_{6,1} \leq 8.7$

☐ Cost of TSP tour is less than or equal to 10 (the bound we placed on the cost of the previous tour using DFS).

☐ There is no information about the cost of the TSP tour due to the missing edge weights.

You didn't select all the correct answers

